



EPILEPSY AND ANTIEPILEPTIC DRUGS

1

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EPILEPSY

- Seizure associated with the episodic high-frequency discharge of impulses by a group of neurons in the brain
- Starts as a local abnormal discharge may then spread to other areas of the brain
- Site of the primary discharge and extent of its spread determine symptoms that are produced



- The particular symptoms produced depend on the function of the region of the brain that is affected
- Involvement of
 - motor cortex- convulsions
 - hypothalamus-peripheral autonomic discharge
 - reticular formation in the upper brain stem - loss of consciousness



TYPES OF EPILEPSY

A. Generalized seizures

- ✓ Generalized tonic-clonic (grand mal) seizures
- ✓ Absence (petit mal) seizures
- ✓ Atonic seizures
- ✓ Clonic and myoclonic seizures
- ✓ Infantile spasm (hypsarrhythmia)



B. Partial seizures

- ✓ Simple partial seizures
- ✓ Complex partial seizures
- ✓ Partial seizures secondarily generalized



A. Generalized seizures(GS)

- Involves the whole brain
- Abnormal electrical activity throughout both hemispheres
- Immediate loss of consciousness



1. Generalized tonic-clonic seizures (grand mal, major epilepsy) - GTCS

- Tonic phase (< 1 min)
 - Sudden loss of consciousness
 - Patient become rigid and falls to ground , respiration arrested
- Clonic phase (2min_s) - Jerking of the body musculature
- Clonic phase followed by prolonged sleep and depression of all CNS functions
- defecation, micturition and salivation often occur



- Immediately after the seizure the patient may
 - recover consciousness
 - drift in to sleep
 - have further convulsion (status epilepticus or serial seizure)
- Convulsion with out recovery of consciousness
(**status epilepticus**)- May lead to brain damage and death
- Further convulsion, after recovering consciousness
(**serial seizure**)



2. Absence (petit mal) seizures— minor epilepsy

- Prevalent in child and cease at the age of 20
- Momentary loss of consciousness, patient freezes and stares in one direction
- Onset and termination of attacks are abrupt
- Patient abruptly ceases whatever he/she was doing
- Impairment of external awareness is so brief that patient is unaware of it
- Many seizures each day may occur as compared to GTCS



3. Atonic seizures (astatic epilepsy)

- Unconsciousness with relaxation of all muscles due to excessive inhibitory discharges

4. Clonic and myoclonic seizures

- Shock like momentary movement, contraction of muscles of a limb or whole body

5. Infantile spasm (hypsarrythmia)

- Intermittent muscle spasm and progressive mental deterioration
- Epileptic syndrome rather than a specific seizure type



B. Partial seizure

- The discharge begins locally and often remains localized
- Symptoms depend on the brain region/regions involved
 - ✓ involuntary muscle contractions
 - ✓ abnormal sensory experiences
 - ✓ autonomic discharge
 - ✓ effects on mood and behaviour (*psychomotor epilepsy*)



1. Simple partial seizures (cortical focal epilepsy)

- Lasts 0.5 – 1 min
- Convolutions are confined to a group of muscles or localized sensory disturbance depends on the area of cortex involved

E.g. If motor cortex supplying to left thumb is affected then jerking of left thumb occurs

- With out loss of consciousness



2. Complex partial seizures (temporal lobe epilepsy, psychomotor)

- Confused behaviors, purposeless movements, emotional changes
- Seizure focus is located in temporal lobe

3. Simple or complex partial seizures secondarily generalized

- Partial seizures occurs followed by GTCS with loss of consciousness



○ Possible causes of seizures

- ✓ primary (idiopathic) – most of the cases
- ✓ secondary to trauma/ surgery on head, intracranial tumor, tuberculoma, cerebral ischemia, etc



MECHANISM OF ACTION OF ANTIEPILEPTIC DRUGS

1. Enhancement of GABA action

- ✓ Barbiturates & BZDs
- ✓ Vigabatrin and valproate (GABA transaminase inhibitor)-increase synaptic GABA conc.
- ✓ Gabapentin (increase GABA release from presynaptic neuron)
- ✓ Tiagabine (inhibit GABA reuptake to presynaptic neuron)

All these facilitate GABA mediated Cl^- channel opening and end up with inhibitory effect



2. Inhibition of Na⁺ channel function

- ✓ Drugs: Phenytoin, Carbamazepine, Valproic acid, Lamotrigine, Topiramate, Zonisamid

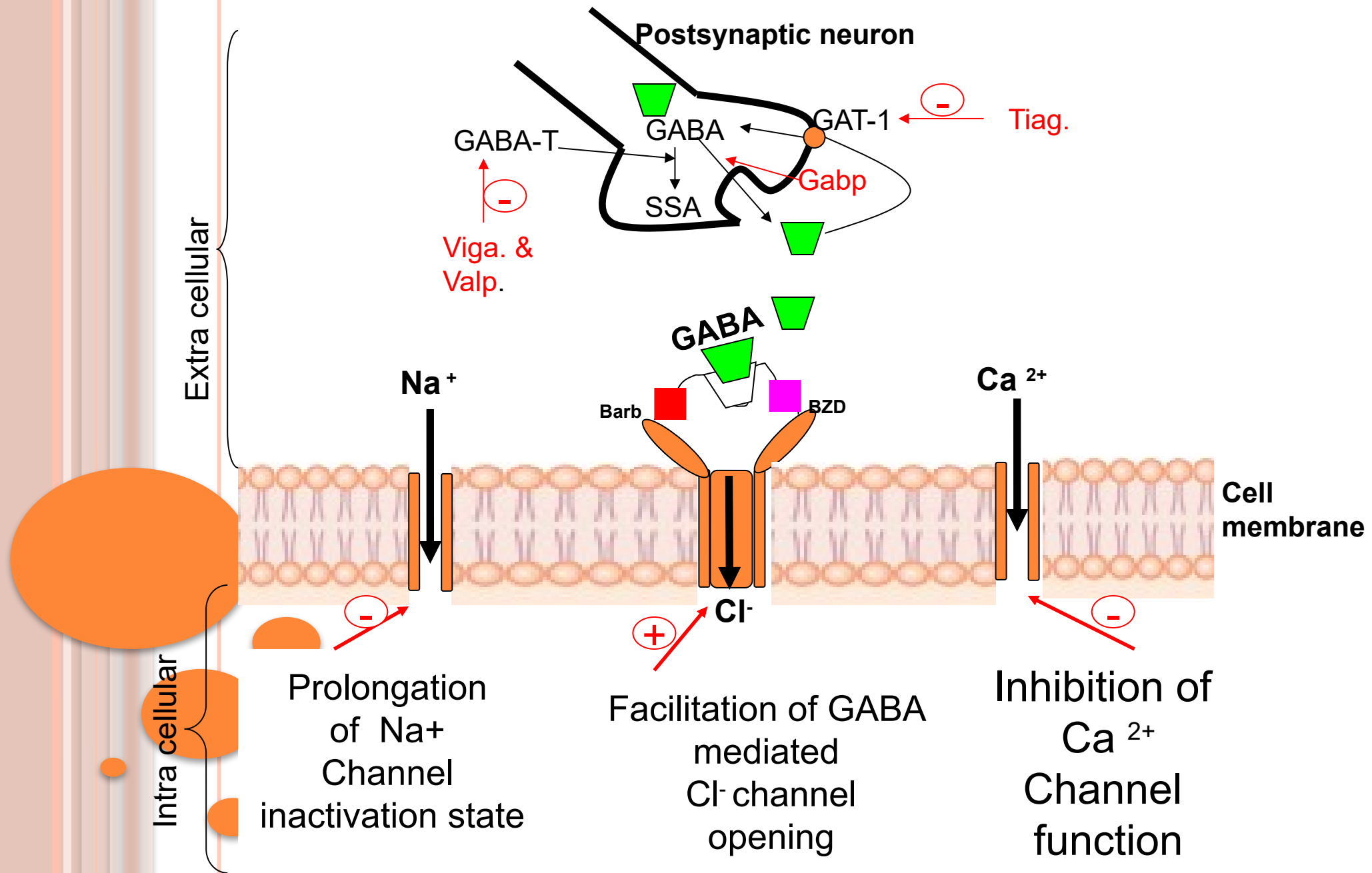
prolong Na⁺channel inactivation state

3. Inhibition of Ca²⁺ channel function

- ✓ Drugs: Ethosuximide, Trimethadione, Valproate

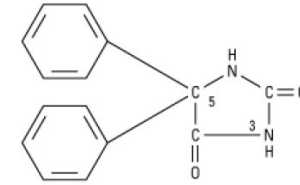
Inhibit T-type Ca²⁺ current





Individual drugs

Phenytoin (diphenylhydantoin)



- Oldest nonsedative antiseizure drug
- Used against partial seizures and generalized tonic-clonic seizures

Mechanism of action

- Prolonging inactive state of voltage sensitive Na^+ channel----- use-dependent effect
- Inhibition of high frequency discharge with little effect on low frequency discharges
- At high/ toxic conc: reduce Ca^{2+} influx, inhibition of glutamate, and facilitate GABA responses



Adverse effects

❖ At therapeutic concentration

- Gum hypertrophy/ gingival hyperplasia (20%)
- Hirsutism, **coarsening of facial feature**, acne
- Megaloblastic anemia(decrease absorption and increased excretion of folate)
- Abnormal Vit D, Vit K and Ca^{+} metabolism (Osteomalacia, Hemorrhage)
- Hyperglycemia / inhibit insulin release
- Fetal hydantoin syndrome (hypoplastic phalanges, cleft palate, microcephaly)



❖ At high plasma level

○ Dose related toxicity

- Ataxia, vertigo, diplopia, nystagmus

- drowsiness, behavioral alterations, mental confusion, hallucination

- Epigastric pain, nausea and vomiting



Drug interaction

- Drug interactions related to plasma protein binding or drug metabolism
- 90% plasma proteins binding
 - Hypoalbuminemia and drugs which displace phenytoin from plasma proteins alter total plasma conc. of the drug
- Phenytoin induce microsomal enzymes
 - Enhance metabolism of drugs which get metabolized by microsomal enzyme

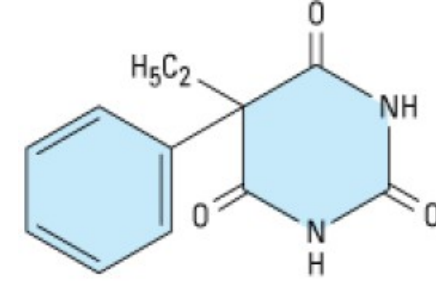


FOSPHENYTON

- Water soluble prodrug of phenytoin
- Introduced to overcome difficulties in i.v. administration of phenytoin in status epilepticus



PHENOBARBITONE



- Effective against GTC, SP and CP seizures
- Can be used for status epilepticus but with slow onset of action
- Ineffective in absence seizure
- less commonly used than phenytoin, carbamazepam, or valproate
- Similar clinical use as phenytoin
- Phenytoin is preferred because of the absence of sedation



Mechanism of action

- Exact mechanism of action is unknown
 - Probably through enhancement of inhibitory processes and diminution of excitatory transmission
- Binds to allosteric regulatory site of GABA_A receptor (inhibitory effect)
 - Enhances the GABA receptor-mediated current by prolonging the openings of the Cl⁻ channels
- Inhibiting excitatory responses by glutamate



Adverse effects

- Sedation
- Tolerance and dependence with prolonged use
- Hypersensitivity
- Nystagmus and ataxia occur at excessive dosage
- Agitation and confusion in the elderly
- Irritability and hyperactivity in children

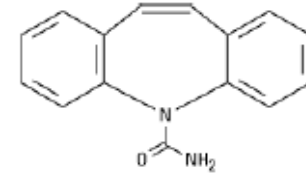


Drug interaction

- Additive effect with CNS depressants
- Induce metabolism of many drugs (warferin, oral contraceptives, chloramphenicol, theophylline, griseofulvin)
- Phenobarbitone competitively inhibits as well as induces phenytoin and imipramine metabolism
- It decrease GIT absorption of griseofulvin
- Plasma conc increased by sodium valproate



Carbamazepine



- Is tricyclic compound and closely related to TCA
- Pharmacological actions resemble phenytoin
- Exert lithium like therapeutic effect in mania and bipolar mood disorder
- Antidiuretic effect / enhance ADH action
- Effective in most forms of epilepsy (except absence seizures)
- Particularly effective in psychomotor epilepsy (CPS)
- Also useful in trigeminal neuralgia



Adverse effects

- Sedation, dizziness, vertigo, diplopia, and ataxia
- Vomiting, diarrhoea
- Worsening of seizure with higher dose
- Water retention and hyponatremia
- Foetal malformation reported
- Teratogenesis doubled if combined with valproate

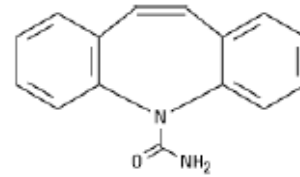


Drug interaction

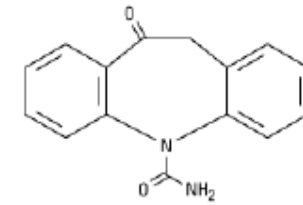
- Enzyme inducer
- Its metabolism induced by phenobarbitone, phenytoin, valproate and vice versa
- Erythromycine, fluoxetine and isoniazid inhibit carbamazepine metabolism



Oxacarbazepine



carbamazepine

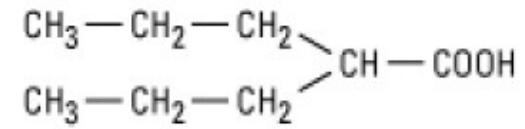


Oxacarbazepine

- Closely related to carbamazepine
- Active metabolite is glucoronide conjugated
- Toxic effect due to epoxide metabolite is avoided
- Useful in the same seizure types but less potent
- Weak enzyme inducer (drug interaction and auto induction of own metabolism are less)
- Risk of hepatotoxicity is less
- Hyponatraemia is more



Valproic acid (sodium valproate)



- Found by serendipity when used as solvent in the search for other drugs effective against seizures
- Effective against GTCS, partial seizures as well as absence seizures
- Ethosuximide is the drug of choice when absence seizures alone occur
- Valproate is better in mixed absence seizures and GTCS
- Mixed absence seizures and GTCS is more common than pure absence seizures



Mechanism of action

- Phenytoin like frequency dependent prolongation of Na^+ channel inactivation
- weak reduction of Ca^{2+} 'T' current (ethosuximide like)
- Effect on GABA metabolism (degradation & uptake inhibition, increase synthesis from glutamic acid)
- Adverse effects (low toxicity)
- Anorexia, vomiting, heart burn, drowsiness, ataxia, tremor
- Spinal bifid and other neural tube defects in the offspring(during pregnancy)



Drug interaction

- Inhibit phenobarbitone metabolism
- Displaces phenytoin plasma protein binding and decrease metabolism (phenytoin toxicity)
- Valproate and carbamazepine induce each other's metabolism
- Fetal abnormality if valproate and carbamazepine given together



Ethosuximide

- Introduced as a "pure petit mal" drug
- Drug used to treat absence seizures and may exacerbate other forms
- Both ethosuximide and valproate are equally effective for absence seizure
- However, valproate is more commonly used

Mechanism of actions : blocking T-type calcium channels

- Relatively few unwanted effects mainly nausea and anorexia



- Phensuxamde and methsuxamide are conegeners
- Methsux. is more toxic and phensux. is less effective

Other drugs

- Topiramate & lamotrigine
- Felbamate, zonisamide & levatiracetum
- *Vigabatrine, Gabapentine & Tiagabine*
- *Oxazolidinediones* (trimethadione, dimethadione & paramethadione)
- BZDs (diazepam & clonazepam)



Drug of choice for.....

1.GTCS/ SP with or with out generalization

1st choice: carbamazepine, phenytoin

2nd choice : valproate, phenobarbitone

2. Complex partial with or with out generalization

1st choice : carbamazepine, valproate, phenytoin

2nd choice: gabapentin, lamotrigine

3. Absence seizure

1st choice: valproate ,

2nd choice: ethosuximide, lamotrigine



4. Status epilepticus

1st choice: diazepam(i.v.), lorazepam(i.v.)

2nd choice: fosphenytoin, phenobarbitone

Alternative/add on drugs : general anesthetics



DRUGS USED FOR THE MANAGEMENT OF PARKINSONISM



INTRODUCTION

- **Parkinsonism** is characterized by a combination of
 - Rigidity,
 - Bradykinesia,
 - Tremor, and
 - Postural instability
- It can occur for a variety of reasons but is usually **idiopathic**



CLINICAL PRESENTATION

Motor Symptoms

- Hypokynesia
- Decreased manual dexterity,
- Diminished arm swing,
- Dysarthria (slurred speech),
- Dysphagia (difficulty with swallowing).

Autonomic and Sensory Symptoms

- The patient experiences **bladder** and anal **sphincter disturbances**
- Constipation
- Diaphoresis



CLINICAL PRESENTATION....

Mental Status Changes

- Anxiety,
- Apathy,
- Bradyphrenia (slowness of thought processes)
- Confusional state,
- Dementia,
- Depression,
- Hallucinosi/psychosis

PATHOGENESIS

- ❖ In normal **conditions** Dopaminergic projections from the SNc to the striatum (putamen and caudate) synapse on two paths
 - To stratum through **direct** and **indirect** pathways
- ❖ which, in turn, mediate motor activity via a complex neuronal circuit involving the **extrapyramidal system**.
- ❖ In Parkinson disease (PD), the degeneration of the SNc neurons results in reduced activity within these two efferent pathways



ETHIOLOGY

- Cause is unknown for most patients.
- It is correlated with destruction of dopaminergic neurons in the substantia nigra
 - Consequently, reduction of dopamine actions in the corpus striatum parts of the brain's basal ganglia system that are involved in motor control.



1. Substantia nigra

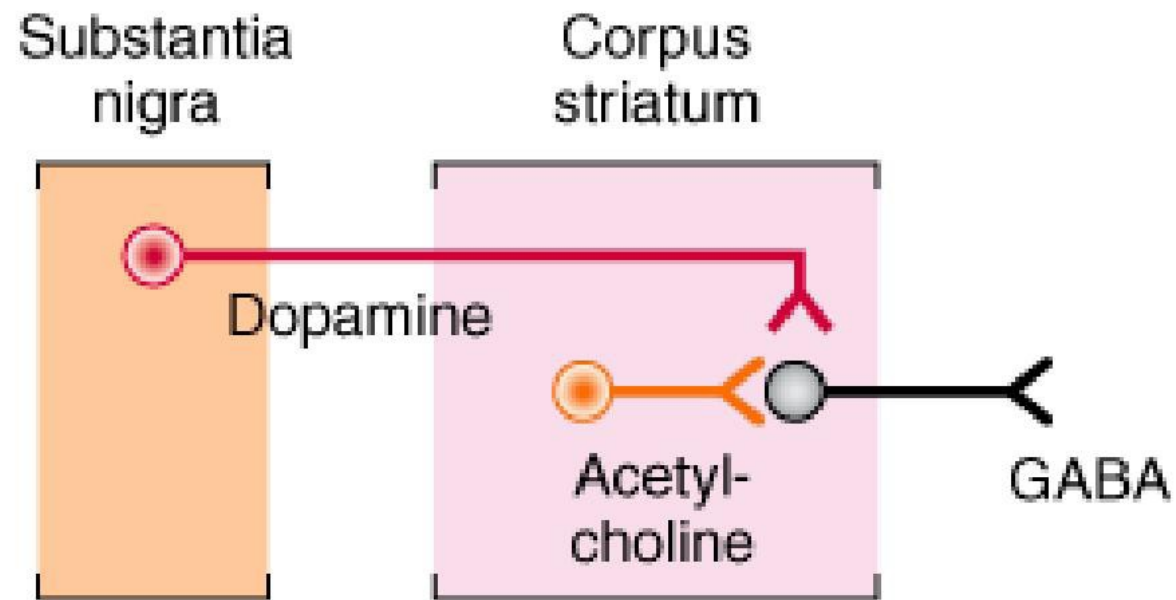
- ▣ Is the source of dopaminergic neurons that terminate in the striatum.
- ▣ Each DAergic neuron synapse with neostriatum
- ▣ Modulates activity of large no. of cells.
- ▣ fire tonically, rather than in response to specific muscular movements or sensory inputs.

2. Neostriatum

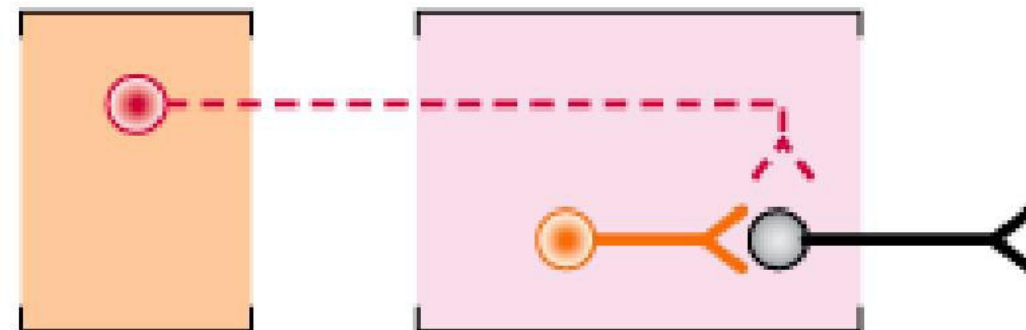
- ▣ Connected to Substantia nigra by neurons that secrete inhibitory NT GABA at their termini in substantia nigra
- ▣ In turn, cells of substantia nigra send neurons back to neostriatum, secreting the inhibitory transmitter Dopamine at their termini.



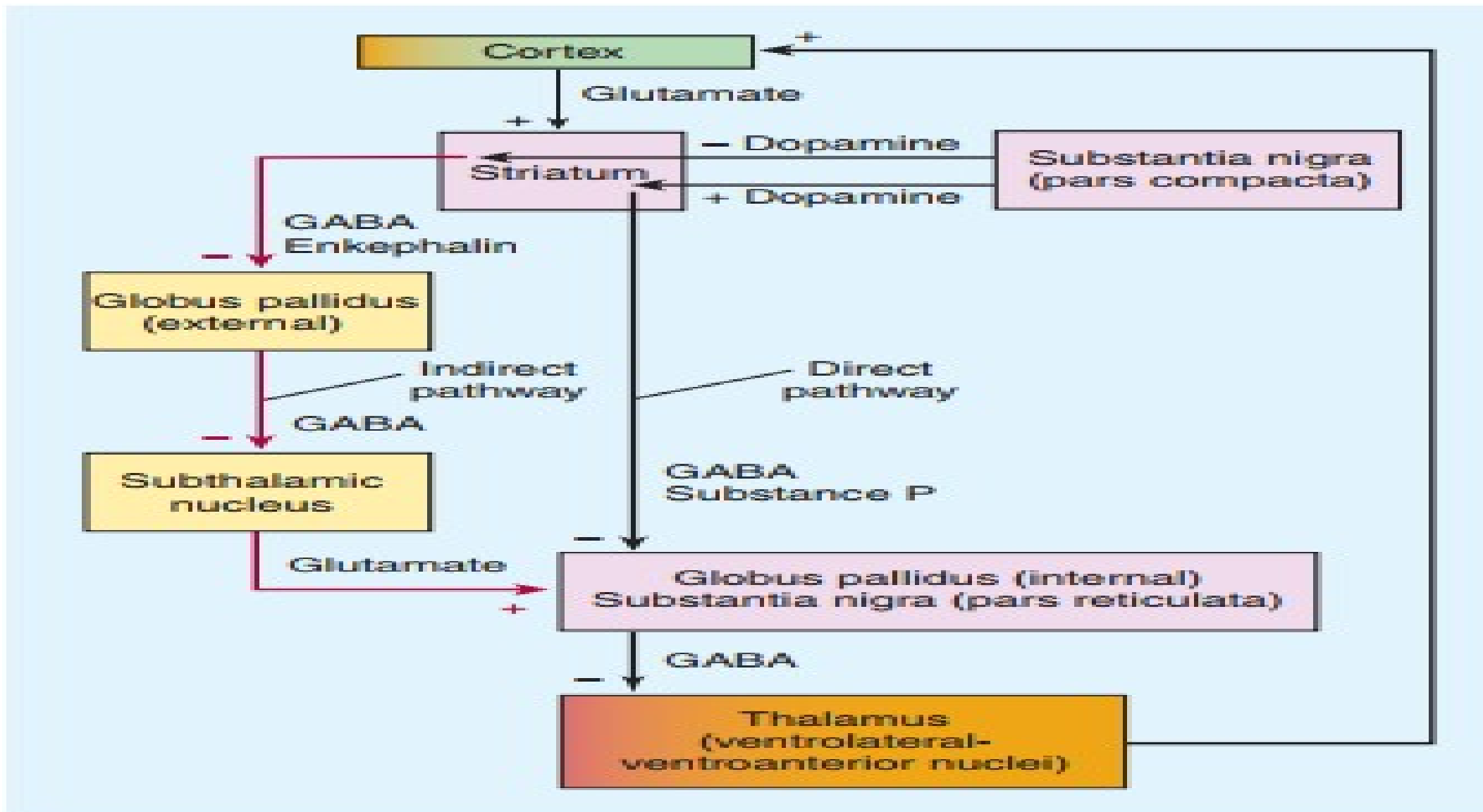
Normal



Parkinsonism

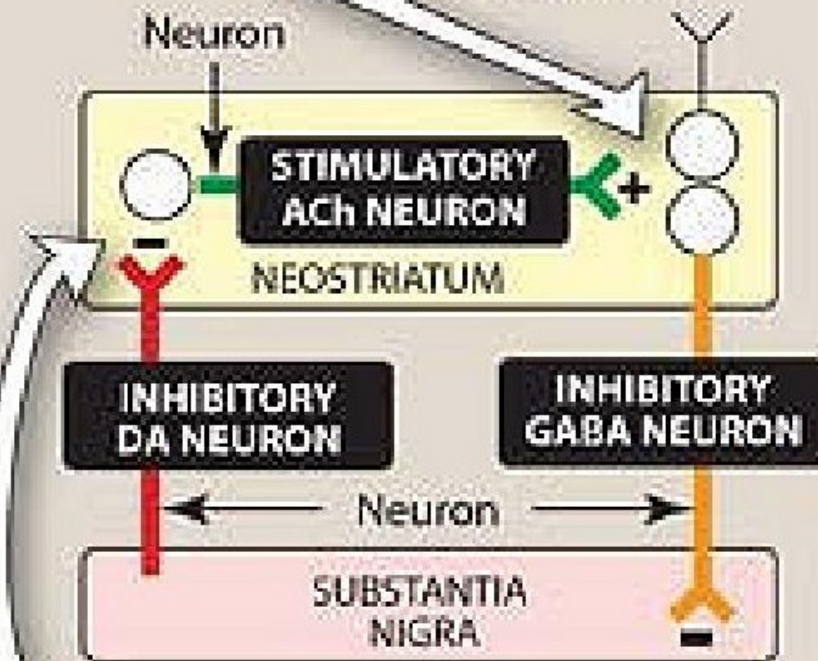


FUNCTIONAL CIRCUITRY BETWEEN THE CORTEX, BASAL GANGLIA, AND THALAMUS.



2 Loss of the inhibitory effect of dopamine results in more production of acetylcholine, which triggers a chain of abnormal signaling leading to impaired mobility.

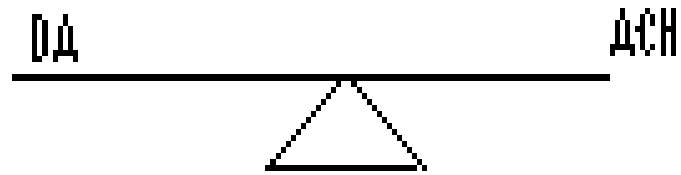
Connections to muscle through motor cortex and spinal chord



1 Cell death results in less dopamine release in the neostriatum.

STRATEGY OF TREATMENT

- Symptoms of Parkinson's reflect
 - imbalance b/n the **excitatory cholinergic neuron** & the greatly diminished number of **inhibitory dopaminergic neurons**.



Normal

Parkinson's disease

- Therapy is aimed
 - to restore the dopamine in the basal ganglia & antagonizing the excitatory effects of cholinergic neurons.
 - Thus re-establishing the correct dopamine and -ACh balance.



ANTICHOLINERGIC MEDICATIONS

- The increased **cholinergic activity** (caused by **dopamine depletion**) is believed to contribute to the **tremor** of **PD**.
- Anticholinergic drugs such as
 - benztropine and
 - Trihexyphenidyl are effective against **tremor**

Levodopa

- Dopamine does not cross the blood-brain but
- levodopa, the immediate metabolic precursor of dopamine, does enter **the brain** (via an L-amino acid transporter, LAT), where it is **decarboxylated** to dopamine



LEVODOPA

Pharmacokinetics

- Levodopa is rapidly absorbed from the small intestine
- certain **amino acids** from ingested food competes for absorption and transportation to CNS
- The main metabolites are
-(homovanillic acid, HVA) and dihydroxyphenylacetic acid (DOPAC).



Pharmacokinetics of Levodopa...

- only about 1–3% enter to CNS without alteration
- remainder is metabolized extra cerebrally (within the periphery) **predominantly** by decarboxylation to **dopamine**.

Therefore

- Give with dopa **decarboxylase inhibitor** (carbidopa)
- Or should be given in large dose if it is alone.

Clinical Use

- Some patients become responsive after 3-4 years
- not stop the progression of parkinsonism but early initiation lowers the mortality rate
- relieving bradykinesia and any disabilities resulting from it



LEVODOPA...

Dosage

- Combination treatment is started with a small carbidopa **25 mg**, levodopa **100 mg** TID.
- It should be taken 30–60 minutes before meals.

Adverse drug reactions

Gastrointestinal Effects

When levodopa is given **without** a peripheral decarboxylase **inhibitor**, **anorexia** and **nausea** and **vomiting** occur in about 80% of patients .to **minimize these ADR**,

- taking the drug in divided doses, with or immediately after meals,
- If dose Increment is requires, increase slowly
- Anti acid administration **30–60 minutes** before levodopa may also be beneficial



LEVODOPA...

Cardiovascular effects –cardiac arrhythmias such as **tachycardia, ventricular extrasystoles, and fibrillation** rarely, **atrial**

- Due increased catecholamine formation peripherally
 - Effects can be decreased by giving carbidopa in combination
 - Postural hypotension is common (but asymptomatic)
 - Hypertension if taken with in a larger dose /if taken with Mao-Inhibitor.

Behavioral Effects - **depression, anxiety, agitation, insomnia, somnolence, sleep attacks, confusion**, delusions, hallucinations, nightmares, euphoria, and other changes in mood or personality



LEVODOPA...

Dyskinesias and Response Fluctuations

- **Dyskinesias** occur in up to 80% of patients receiving levodopa therapy for more than 10 years.

Drug Holidays to reduce some of the Adverse effects

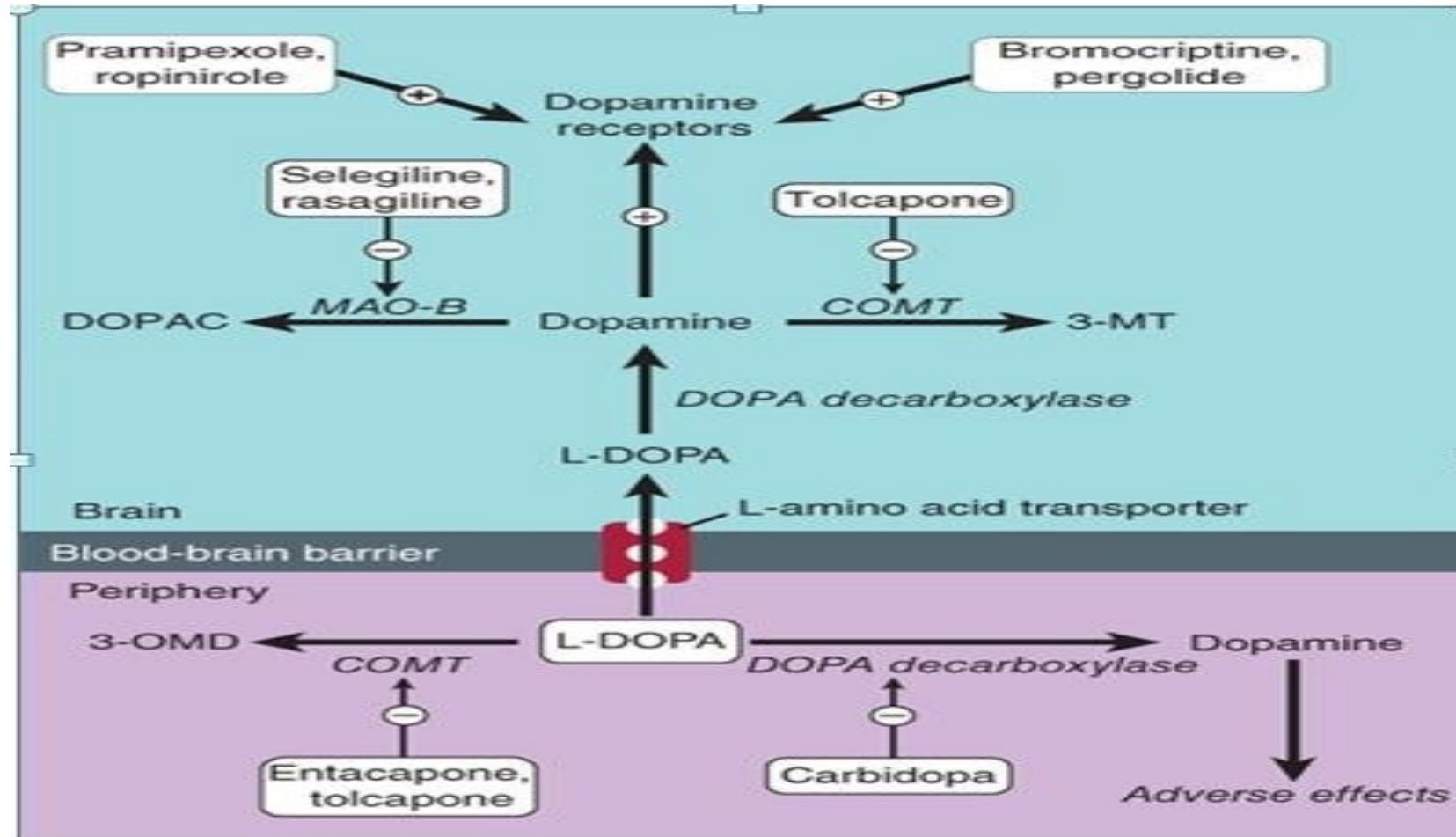
- A drug holiday (discontinuance of the drug for 3–21 days) may temporarily improve **responsiveness** to levodopa and alleviate some of its **adverse effects**

Drug interaction

- Pharmacologic doses of pyridoxine (vitamin B₆) enhance the extracerebral metabolism of levodopa...reduces its effects.
- Levodopa should not be given to patients taking monoamine oxidase can lead to **hypertensive crises**.



DOPAMINE RECEPTOR AGONISTS AND OTHERS



DOPAMINE RECEPTOR AGONISTS

- ❖ Drugs acting directly on **postsynaptic dopamine receptors**
 - may have a **beneficial effect** compared to **levodopa**
 - They do not require **enzymatic conversion** to an active metabolite,
 - Have no potentially **toxic metabolites**, and
 - Do not compete with other **substances for active transport** into the blood and across the blood-brain barrier.
 - May have more **limited adverse effects** than levodopa
- ❖ **Drugs:- Ergots** (Bromocriptine, Pergolide), Pramipexole, Ropinirole, Rotigotine



DOPAMINE RECEPTOR AGONISTS

Bromocriptine

- Bromocriptine is a D₂ agonist;
- Rarely used currently due to the advent of newer **DRA**s
- The usual daily dose varies between **7.5** and **30** mg.
- To minimize adverse effects, the dose is built up slowly over 2 or 3 months.

Pergolide

- Stimulates both **D₁** and **D₂** receptors.
- Associated with the development of valvular heart disease...Not used in **U.S.A**



DOPAMINE RECEPTOR AGONISTS...

Pramipexole

- it has preferential affinity for the D3 family of receptors.
- It is effective as monotherapy for mild and helpful in patients with advanced disease
- may ameliorate affective symptoms.
- A possible neuroprotective effect has been suggested.
- **Dose 0.125 TID**, doubled after 1 week, and again after **2 weeks** Further increments in the daily dose by **0.75 mg** at weekly intervals, depending on response and tolerance.
- Most patients require between **0.5 and 1.5 mg TID**



DOPAMINE RECEPTOR AGONISTS...

Ropinirole

- is a relatively pure D₂ receptor agonist.
- It is introduced at **0.25** (TID)....Increased by **0.75 mg** weekly until the **fourth week** and by **1.5 mg** thereafter.
- In **most instances**, a dosage between 2 and 8 mg three times daily is necessary.
- Ropinirole is metabolized by CYP1A2.

Rotigotine

- delivered daily through a **skin patch**
- reactions may also at the application site...may be serious



ADVERSE EFFECTS OF DOPAMINE AGONISTS

Gastrointestinal effects

- Anorexia and nausea and vomitingcan be minimized by taking the medication with meals.
- Constipation, dyspepsia, and symptoms of reflux esophagitis
- Bleeding from peptic ulceration

Cardiovascular effects

- Postural hypotension during **initiation of therapy**.
- **Painless digital vasospasm** with long use of ergot derivative
- **cardiac arrhythmias**need discontinuing treatment.
- **Peripheral edema**.
- Cardiac valvulopathy may occur with **pergolide**.



ADVERSE EFFECTS OF DOPAMINE AGONISTS...

Dyskinesias

- Abnormal movements similar to those introduced by levodopa
 - reversed by reducing the total dose of dopaminergic drugs being taken.

Mental disturbances

- Confusion, hallucinations, delusions, and other psychiatric reactions as a feature of **Parkinson's disease**
 - more common and severe than with levodopa
 - Occurs in older with no difference in scale of toxicity among this class of drugs
- They may respond to atypical antipsychotic agents such as **clozapine, olanzapine, quetiapine, and risperidone** or to **pimavanserin**.



DOPAMINE RECEPTOR AGONISTS...

Contraindications

- Psychotic illness or
- Recent myocardial infarction,
- Active peptic ulcer.
- The ergot-derived agonists are best avoided in patients with peripheral vascular disease.



MONOAMINE OXIDASE INHIBITORS

- ❖ There are 2 **types of monoamine oxidase (MAO)** in the nervous system.
 - **MAO A** metabolizes norepinephrine, serotonin, and dopamine; Where
 - as
 - **MAO B** metabolizes dopamine selectively.
- ❖ Drugs that inhibits this enzyme...**selegiline, rasagiline, Saffinamide**
- **Selegiline**
- ❖ a selective irreversible inhibitor of **MAO B** (at normal/higher dose) it inhibits **MAO-A** as well).
 - prolongs the antiparkinsonism effect of levodopa (thereby
 - allowing the dose of levodopa to be reduced).
 - But has **minimal effects** when it is given alone.



SELEGILINE....

- Used as adjunctive therapy for patients with a declining or **fluctuating** response to **levodopa**
 - **Dose =5 mg** with breakfast & with lunch.
 - Selegiline may cause insomnia
- **Rasagiline,**
- MAO-B inhibitor, which is more potent than **selegiline** in preventing (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine)MPTP induced parkinsonism and
 - used for early treatment in patients with mild symptoms.
 - The standard dosage is 1 mg/d.
- **Rasagiline** used as adjunctive therapy at a dosage of **0.5** or **1 mg/d** to prolong the effects of **carbidopa-levodopa** in patients with **advanced disease** and **response fluctuations**.



SAFINAMIDE

- **It is a MAO-B inhibitors**
- It is not effective as monotherapy for Parkinson's disease but It is used to reduce
 - response fluctuations in patients taking carbidopa-levodopa and diminishing **off-periods** in patients with wearing-off effect or on-off phenomena.
- . The starting dose is **50 mg** orally once daily, increased after 2 weeks to 100 mg once daily.



CATECHOL-*O*-METHYLTRANSFERASE INHIBITORS

- **Drugs....tolcapone and entacapone**
 - prolong the action of levodopa by diminishing its peripheral metabolism
 - Inhibits 3-*O*-methyldopa (3-OMD) effects on the transporter of levodopa
 - Reduce total daily levodopa dose
- **Entacapone** is generally preferred because it has not been associated with **hepatotoxicity**.



- ❖ The pharmacologic effects of tolcapone and entacapone are similar,
 - both are rapidly absorbed, bound to plasma proteins, and metabolized before excretion.
 - However,
 - **tolcapone** has both **central** and **peripheral effects**, whereas the effect of entacapone is peripheral.
 - **tolcapone** is slightly more potent and has a longer duration of action.

Dose=

- **Tolcapone** is taken in a standard dosage of **100 mg** three times
- **entacapone (200 mg)** needs to be taken with each dose of levodopa, up to six times daily.



CATECHOL-*O*-METHYLTRANSFERASE INHIBITORS

- ❖ Adverse effects of the COMT inhibitors relate in part to increased levodopa.
 - It is often necessary to lower the daily dose of levodopa by about **30%** in the first **48 hours** to avoid or reverse such complications.
- ❖ Other adverse effects include
 - diarrhea,
 - abdominal pain,
 - orthostatic hypotension,
 - sleep disturbances, and an
 - orange discoloration of the urine.
 - Hepatotoxicity



APOMORPHINE

MOA-

interacts with postsynaptic **D₂** receptors in the **caudate nucleus** and **putamen**,

- is effective for the temporary **relief** (“rescue”) of **off-periods** of akinesia in patients on optimized dopaminergic therapy.
- It is rapidly taken up in the blood

PK

- Onset of action = **10 minutes**
- Duration = **2 hours**.

Dose = Most patients require a dose of 0.3–0.6 mL (3–6 mg), and this should be given usually no more than about three times daily, but occasionally up to **five** times daily.



APOMORPHINE...

Adverse drug reactions

- ❖ Nausea, especially at the initiation of apomorphine treatment;
 - Needs pre-treatment with trimethobenzamide (**300 mg** TID daily) for **3** days and is then continued for at least **1** month,
- ❖ Others adverse effects include
 - dyskinesias, drowsiness, insomnia,
 - chest pain, sweating, hypotension,
 - syncope, constipation, Diarrhea,
 - bruising at the injection site.

Apomorphine It should not be used in patients taking serotonin **5-HT₃** antagonists because severe hypotension may result.



AMANTADINE

- ❖ Amantadine, an antiviral agent, was
- ❖ Found to have antiparkinson effect by chance
- ❖ **MOA**
 - Its mode of action in parkinsonism is unclear, but it may potentiate dopaminergic function through
 - influencing the synthesis, release, or reuptake of dopamine.
 - It has been reported to antagonize the effects of adenosine at adenosine **A2A** receptors, which may inhibit D2 receptor function..
 - Amantadine is an antagonist of the **NMDA**-type glutamate receptor, suggesting an antidyskinetic effect.
 - Dose = **100 mg** orally **two** or **three** times daily



THERAPY FOR NONMOTOR MANIFESTATIONS

- ❖ cognitive decline may respond to
 - rivastigmine (1.5–6 mg twice daily),
 - memantine (5–10 mg daily), or
 - donepezil (5–10 mg daily)
- ❖ with excessive daytime sleepiness to modafinil
 - 100–400 mg in the morning
- ❖ bladder and bowel disorders
 - appropriate symptomatic therapy

